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Chapter 1 Groups and Homomorphisms

Exercise 1

Let N be a subgroup of G and $g \in G$. Then $g^{-1}Ng = \{g^{-1}ng \mid n \in N\} = \{g^{-1}gn \mid n \in N\} = \{n \mid n \in N\} = N$ because G is abelian. So any subgroup of G is normal. Since G is simple, the only subgroups of G are $\{1\}$ and G . Also, using the fact that G is simple we can assume that $G \neq \{1\}$. Choose $h \in G \setminus \{1\}$. Since $\langle h \rangle$ is a subgroup of G , $\langle h \rangle = G$ or $\langle h \rangle = \{1\}$. If $\langle h \rangle = \{1\}$, then $h = 1$, a contradiction. Hence, $\langle h \rangle = G$, so G is a cyclic group. Let $|G| = n$. Assume for contradiction that n is not prime, so $n = ab$ for some $1 < a, b < n$. Since G is cyclic and a is a factor of n , there exists $k \in G$ of order a . Since $|k| > 1$, $k \neq 1$. By the above argument, we have $\langle k \rangle = G$. Therefore, $a = |k| = |\langle k \rangle| = |G| = n$, a contradiction. Hence, G is cyclic of prime order. $\square$

Exercise 2

Assume $H \neq \{1\}$. Let $g_1, g_2 \in G$. Suppose $g_1\vartheta = g_2\vartheta$. Then $(g_1\vartheta)(g_2\vartheta)^{-1} = 1$ and thus $(g_1g_2^{-1})\vartheta = 1$. Hence $g_1g_2^{-1} \in \text{Ker}(\vartheta)$. Since $\text{Ker}(\vartheta) \triangleleft G$, we have $\text{Ker}(\vartheta) = \{1\}$ or $\text{Ker}(\vartheta) = G$. If $\text{Ker}(\vartheta) = G$, then $g\vartheta = 1$ for all $g \in G$. Since ϑ is surjective, $H = \text{Im}(\vartheta) = \{g\vartheta \mid g \in G\} = \{1\}$, a contradiction. Therefore, $\text{Ker}(\vartheta) = \{1\}$. Hence, we have $g_1g_2^{-1} = 1$, which implies $g_1 = g_2$. So ϑ is injective and thus is bijective. $\square$

Chapter 2 Vector Spaces and Linear Transformations

Exercise 3

Suppose $V = U \oplus W$. Clearly, $V = U + W$. Let $v \in U \cap W \subseteq V$. Then there exist unique $u \in U$ and $w \in W$ such that $v = u + w$. So $u + w \in U$ and $u + w \in W$. By the uniqueness of u and w , we have $u + w = u$ and $u + w = w$, so $u = w = 0$, and thus $v = 0$. Hence, $U \cap W = \{0\}$.

Now suppose $V = U + W$ and $U \cap W = \{0\}$. We want to prove that for any $v \in V$ there exist unique $u \in U$ and $w \in W$ such that $v = u + w$. Let $v \in V$. Suppose $u_1, u_2 \in U$ and $w_1, w_2 \in W$ satisfy $v = u_1 + w_1 = u_2 + w_2$. Then $u_1 - u_2 = w_2 - w_1$. Since U and W are subspaces, they are closed under vector addition and additive inverses, so $u_1 - u_2 \in U$ and $w_2 - w_1 \in W$, so $u_1 - u_2 = w_2 - w_1 \in U \cap W$. Hence, $u_1 - u_2 = w_2 - w_1 = 0$, and thus $u_1 = u_2$ and $w_1 = w_2$, proving the uniqueness of u and w .

Therefore, $V = U \oplus W$ if and only if $V = U + W$ and $U \cap W = \{0\}$. $\square$

Exercise 4

Suppose $V = U \oplus W$. Let $v \in V$. Then there exist unique $u \in U$ and $w \in W$ such that $v = u + w$. Since $u_1, \dots, u_r$ is a basis of U , there exist unique $\lambda_1, \dots, \lambda_r \in F$ such that $u = \lambda_1 u_1 + \dots + \lambda_r u_r$. Since $w_1, \dots, w_s$ is a basis of W , there exist unique $\mu_1, \dots, \mu_s \in F$ such that $w = \mu_1 w_1 + \dots + \mu_s w_s$. Therefore, there exist unique $\lambda_1, \dots, \lambda_r, \mu_1, \dots, \mu_s \in F$ such that $v = \lambda_1 u_1 + \dots + \lambda_r u_r + \mu_1 w_1 + \dots + \mu_s w_s$. Hence, $u_1, \dots, u_r, w_1, \dots, w_s$ is a basis of V .

Now suppose $u_1, \dots, u_r, w_1, \dots, w_s$ is a basis of V . Let $v \in V$. Then there exist unique $\lambda_1, \dots, \lambda_r, \mu_1, \dots, \mu_s \in F$ such that $v = \lambda_1 u_1 + \dots + \lambda_r u_r + \mu_1 w_1 + \dots + \mu_s w_s$. Since $u_1, \dots, u_r$ is a basis of U , there exists a unique $u \in U$ such that $u = \lambda_1 u_1 + \dots + \lambda_r u_r$. Since $w_1, \dots, w_s$ is a basis of W , there exists a unique $w \in W$ such that $w = \mu_1 w_1 + \dots + \mu_s w_s$. Hence, there exist unique $u \in U$ and $w \in W$ such that $v = u + w$. So $V = U \oplus W$.

Therefore, $V = U \oplus W$ if and only if $u_1, \dots, u_r, w_1, \dots, w_s$ is a basis of V . $\square$